generic_light_gbm Â¶

GenericLightGBMConfig Â¶

```
GenericLightGBMConfig(builder, boosting_type: str = 'gbdt', num_iterations: int = 100, learning_rate: float = 0.1, num_leaves: int = 31, lightgbm_seed: int = 0, max_depth: int | None = None, min_data_in_leaf: int = 20, min_sum_hessian_in_leaf: float = 0.001, bagging_fraction: float = 1.0, bagging_freq: int = 0, feature_fraction: float = 1.0, feature_fraction_bynode: float = 1.0, lambda_l1: float = 0.0, lambda_l2: float = 0.0, min_gain_to_split: float = 0.0, drop_rate: float = 0.1, max_drop: int = 50, skip_drop: float = 0.5, xgboost_dart_mode: bool = False, uniform_drop: bool = False, top_rate: float = 0.2, other_rate: float = 0.1, min_data_per_group: int = 100, max_cat_threshold: int = 32, cat_l2: float = 10.0, cat_smooth: float = 10.0, max_cat_to_onehot: int = 4, verbosity: int = 1, max_bin: int = 255, min_data_in_bin: int = 3, bin_construct_sample_cnt: int = 200000, enable_bundle: bool = True, zero_as_missing: bool = False, is_unbalance: bool = False, undersampling: float | None = None, oversampling: float | None = None, seed: int | None = None, ignored_fields: set[str] | None = None, schema: Schema | None = None)
```

Bases: `TrainingConfig`

Generic GBM configuration object. Serves as parent class for LightGBMConfig and FairGBMConfig.

Parameters:

Name	Type	Description	Default
`builder` Â¶		Builder that comes from the specific algorithm.	*required*
`boosting_type` Â¶	`str`	Boosting type.	`'gbdt'`
`num_iterations` Â¶	`int`	Number of booster iterations.	`100`
`learning_rate` Â¶	`float`	Learning rate.	`0.1`
`num_leaves` Â¶	`int`	Maximum tree leaves.	`31`
`lightgbm_seed` Â¶	`int`	Seed used to generate other seeds.	`0`
`max_depth` Â¶	`int | None`	Maximum tree depth.	`None`
`min_data_in_leaf` Â¶	`int`	Minimum leaf data points.	`20`

Name	Type	Description	Default
`min_sum_hessian_in_leaf` Â¶	`float`	Minimal sum of hessian per leaf.	`0.001`
`bagging_fraction` Â¶	`float`	Bagging sample fraction.	`1.0`
`bagging_freq` Â¶	`int`	Bagging frequency.	`0`
`feature_fraction` Â¶	`float`	Feature fraction by tree.	`1.0`
`feature_fraction_bynode` Â¶	`float`	Feature fraction by tree node.	`1.0`
`lambda_l1` Â¶	`float`	L1 regularization.	`0.0`
`lambda_l2` Â¶	`float`	L2 regularization.	`0.0`
`min_gain_to_split` Â¶	`float`	Minimal gain to split a leaf.	`0.0`
`drop_rate` Â¶	`float`	DART drop rate.	`0.1`
`max_drop` Â¶	`int`	DART max drop.	`50`
`skip_drop` Â¶	`float`	DART skip drop.	`0.5`
`xgboost_dart_mode` Â¶	`bool`	DART XGBoost mode.	`False`
`uniform_drop` Â¶	`bool`	DART uniform drop.	`False`
`top_rate` Â¶	`float`	GOSS top retain rate.	`0.2`
`other_rate` Â¶	`float`	GOSS other retain rate.	`0.1`
	`int`	Minimum data per group.	`100`

Name	Type	Description	Default
`min_data_per_group` Â¶			
`max_cat_threshold` Â¶	`int`	Maximum categories thresholds.	`32`
`cat_l2` Â¶	`float`	Categorical L2 regularization.	`10.0`
`cat_smooth` Â¶	`float`	Categorical smoothing.	`10.0`
`max_cat_to_onehot` Â¶	`int`	Max categories to one-hot encode.	`4`
`verbosity` Â¶	`int`	Verbosity of the LightGBM backend.	`1`
`max_bin` Â¶	`int`	Max bins.	`255`
`min_data_in_bin` Â¶	`int`	Minimum bin data.	`3`
`bin_construct_sample_cnt` Â¶	`int`	Bin construction sample count.	`200000`
`enable_bundle` Â¶	`bool`	Exclusive Feature Bundling (EFB).	`True`
`zero_as_missing` Â¶	`bool`	Use zeroes as missing.	`False`
`is_unbalance` Â¶	`bool`	Unbalanced label.	`False`
`undersampling` Â¶	`float`	The value for undersampling. Only this or oversampling can be used.	`None`
`oversampling` Â¶	`float`	The value for oversampling. Only this or undersampling can be used.	`None`
`seed` Â¶	`int`	The seed value.	`None`

Name	Type	Description	Default
ignored_fields Â¶	set of str	Set of names corresponding to the fields to be ignored when training a model.	None
schema Â¶	Schema	Pass a schema to enable conversion between descs and internal names. By not passing a schema, ds-api will assume field names are references to internal Pulse names, not the ones in the UI.	None

bagging_fraction property writable Â¶

bagging_fraction

bagging_freq property writable Â¶

bagging_freq

bin_construct_sample_cnt property writable Â¶

bin_construct_sample_cnt

boosting_type property writable Â¶

boosting_type

cat_l2 property writable Â¶

cat_l2

cat_smooth property writable Â¶

cat_smooth

drop_rate property writable Â¶

drop_rate

enable_bundle property writable Â¶

enable_bundle

feature_fraction property writable Â¶

feature_fraction

feature_fraction_bynode property writable Â¶

feature_fraction_bynode

ignored_fields property writable Â¶

ignored_fields

is_unbalance property writable Â¶

is_unbalance

lambda_l1 `property` `writable` Â¶

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lambda_l1
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lambda_l2 `property` `writable` Â¶

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lambda_l2
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learning_rate `property` `writable` Â¶

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learning_rate
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lightgbm_seed `property` `writable` Â¶

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lightgbm_seed
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max_bin `property` `writable` Â¶

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max_bin
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max_cat_threshold `property` `writable` Â¶

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max_cat_threshold
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max_cat_to_onehot `property` `writable` Â¶

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max_cat_to_onehot
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max_depth `property` `writable` Â¶

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max_depth
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max_drop `property` `writable` Â¶

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max_drop
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min_data_in_bin `property` `writable` Â¶

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min_data_in_bin
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min_data_in_leaf `property` `writable` Â¶

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min_data_in_leaf
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min_data_per_group `property` `writable` Â¶

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min_data_per_group
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min_gain_to_split `property` `writable` Â¶

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min_gain_to_split
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min_sum_hessian_in_leaf `property` `writable` Â¶

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min_sum_hessian_in_leaf
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num_iterations `property` `writable` Â¶

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num_iterations
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num_leaves `property` `writable` Â¶

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num_leaves
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other_rate `property` `writable` Â¶

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other_rate
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oversampling `property` `writable` Â¶

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oversampling
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seed `property` `writable` Â¶

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seed
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skip_drop `property` `writable` Â¶

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skip_drop
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subclasses `class-attribute` `instance-attribute` Â¶

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subclasses: List[JavaWrapper] = []
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top_rate `property` `writable` Â¶

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top_rate
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undersampling `property` `writable` Â¶

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undersampling
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uniform_drop `property` `writable` Â¶

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uniform_drop
```

verbosity `property` `writable` Â¶

```
verbosity
```

xgboost_dart_mode `property` `writable` Â¶

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xgboost_dart_mode
```

zero_as_missing `property` `writable` Â¶

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zero_as_missing
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